Open Rack, Liquid Cooling… Now what?
Operationalizing the Next Gen Datacenter

Joseph “JBK” Knight, Chief Technologist - Service Providers

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Joseph “JBK” Knight

Chief Technologist - Service Providers

Agenda

01 The Emerging Operations Gap

02 The Rack is the Computer

03 HPE is Building on Proven Foundations

OCP Solved Infrastructure, Who Solves Operations?

The industry is shifting from server-centric management to rack-centric operations

What OCP Successfully Changed

- Rack & Power introduced shared power architectures and rack-level power domains.
- Cooling Environments for liquid infrastructure, CDUs, manifolds, and thermal management beyond the server boundary.
- DC-MHS disaggregated platform components.

The Operational Gap

- How do infrastructure and servers communicate?
- How is rack infrastructure discovered?
- How are power sequencing and policies managed?
- Is different or additional instrumentation needed?
- How is telemetry synthesized and reported, and appropriate actions triggered?

The Rack is the Computer

The industry is shifting from server-centric management to rack-centric operations

Platform vendors should deliver . . .

- Discover
- Telemetry
- Fault isolation
- Lifecyle management
- Power management
- Cooling management

. . . At rack-scale

- The rack becomes the operational unit
- Infrastructure becomes self-describing
- Health is visible beyond the server
- Power, cooling, and compute act as one system
- Customers define policy, not integration



HPE Is Building on Proven Foundations

- **iLO established a server-centric operational model** for discovery, health, lifecycle management, and automation.
- **APM and PCIM extends management beyond the server** into rack power, thermal infrastructure, and liquid-cooling environments.
- **Rack-scale infrastructure demands new management hierarchies** spanning systems, chassis, rack controllers, and infrastructure domains.
- **Power, cooling, and compute can no longer be managed independently** as rack-scale policies increasingly span all three domains.
- **Open interfaces and standards provide the integration path** across servers, rack controllers, power systems, and cooling infrastructure.
- **The goal is a consistent operational experience** across ProLiant and HPC/AI, Open Rack and EIA 19”, DC-MHS and MGX



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Thank you!

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jbk@hpe.com